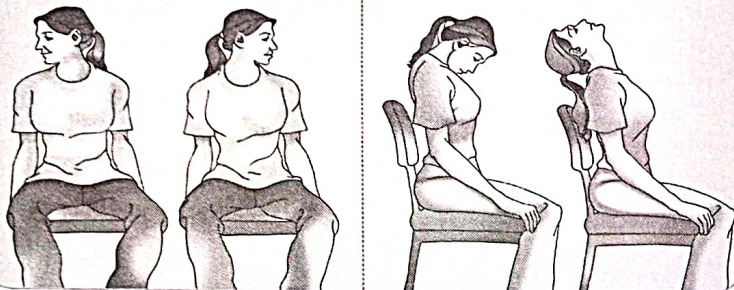


Neck & Shoulder Exercises

These exercises strengthen your neck and shoulders. Before beginning any exercises, please read all the instructions carefully and consult your doctor. While exercising, breathe normally and use smooth movements. If you feel any pain, stop the exercise and consult your doctor. You can use a chair with a good back support to do sitting exercises.

Neck Rotation & Extension

1

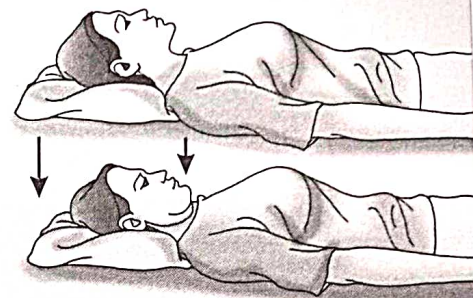


Slowly turn your head to the right as far as possible. Then, slowly turn towards your left. Relax and repeat as advised.

Slowly lower your chin towards chest, keeping your mouth closed. Go back to a neutral position. Tilt your head back very slowly as far as you are comfortable to look up at the ceiling. Bring your neck to a neutral position and relax. Repeat as advised.

Chin Tuck

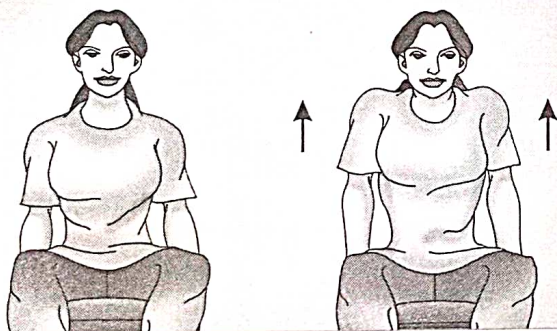
2



Lie flat on your back. Place a thin pillow under your head. Tuck in your chin and push your head and neck into the pillow. Hold for a few seconds and relax. Repeat as advised.

Shoulder Shrugs

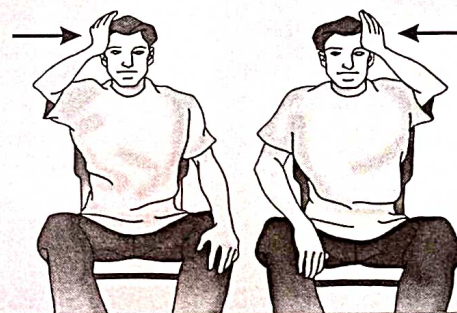
3



Sit on a chair and keep your palms at the back of your head. Bring your elbows forward and take them back in alignment with your shoulders. Repeat as advised and relax.

Neck Resistance

4

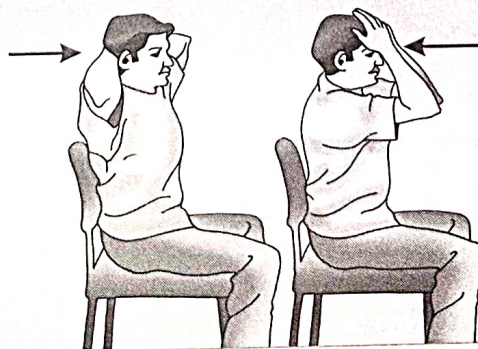


Sit on a chair. Press your right palm on the right side of your head. Push your head against your palm without allowing your head to move and hold for a few seconds.

Press your left palm on the left side of your head. Push your head against your palm without allowing your head to move and hold for a few seconds.

Neck Resistance

5



Take both your palms at the back of your head. Press your head against your palms, so that your head doesn't move and hold for a few seconds.

Place both your palms on your forehead. Press your head into your palms while not allowing your head to move and hold for a few seconds.

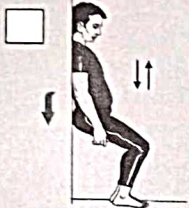
Back Exercises

Follow these simple exercises for your back



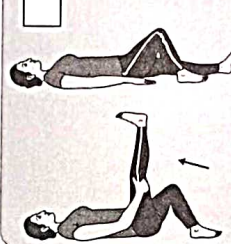
Standing Arch

Stand with your feet apart and hands on your back with fingers pointing backwards. Bend backwards at the waist, supporting the trunk with your hands. Keep your knees straight. Hold for a count of 5. Repeat this 10 times.



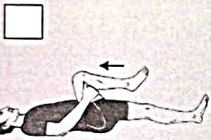
Wall Slides

Stand with back leaning against a wall. Bring your feet one foot forward to the wall. Keep your abdominal muscles tight and slowly bend your knees. Hold for a few seconds and slowly return to the original position. Repeat this 10 times.



Hamstring Stretch

Lie on your back with one leg bent and your other leg extended straight out. Slowly lift your straight leg upward until you feel a slight stretch along the back of your leg. Use your hands to grasp behind your knee and help raise and hold your leg. Hold for a count of 5 and then return to the original position. Repeat with alternate legs.



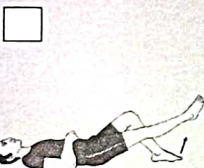
Single Knee to Chest

Lie on your back with one leg bent, another leg flat on the floor. Pull the bent knee towards your chest while keeping your lower back and other knee pressed against the floor. Hold for a few seconds. Relax. Repeat with alternate legs.



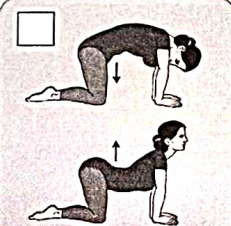
Double Knee to Chest

Grasp your knees with your hands and pull them towards your chest. Hold for a few seconds and relax. Let your knees return, but keep them bent at arm's length. Repeat this 10 times.



Straight Leg Raises

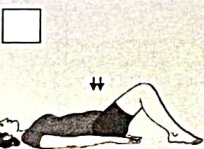
Lie on your back with one leg straight and one knee bent. Tighten abdominal muscles to stabilize low back. Slowly lift one leg straight up about 6 to 12 inches and hold for a count of 5. Slowly return to the original position. Repeat with alternate legs for 10 times.



Cat and Camel

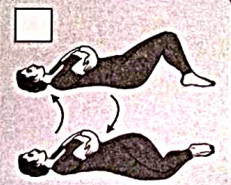
Kneel down on the floor on your hands and knees. Curl your back up towards the ceiling like a cat. Hold for a count of five. Return to starting position.

Pull your stomach down to the floor hollowing out your back like a camel. Hold for a count of five. Return to starting position. Repeat ten times.



Pelvis Tilt

Lie on your back. Bend your knees and keep the feet flat on the surface. Push your back against the floor by pulling the lower abdominal muscles up and in. Hold your back flat while breathing easily in and out. Hold for a count of five. Do not hold breath.



Hip Rolling

Lie on your back. Bend your knees and keep the feet flat on the surface. Cross your arms over your chest. Turn your head to one direction while turning both your knees to the opposite direction. Return to the original position and relax. Now repeat in the reverse direction. Repeat with alternate sides.



Back Extension

Lie on your stomach. Place your arms beside your hips. Raise your head and shoulders off the ground as high as comfortably possible. Hold for a few seconds. Lower the head and shoulders. Do not tense your shoulder muscles. Repeat for few times and relax.



Press Ups

Lie on your stomach. Use your arms to press up. Concentrate on keeping your hips down and push up your upper body as high as possible.

Do's

- ✓ Use a firm mattress.
- ✓ Sleep on your back with a pillow under your thighs or, sleep on your side with your knees bent and a pillow between your legs.
- ✓ Have your work surface at a comfortable height.
- ✓ Bend your knees as you lift. Make your legs do the hard work, not your back.
- ✓ Sit with a rolled up towel or some other support behind your lower back.

Dont's

- ✗ Lift objects away from your body with your arms outstretched or objects that are too heavy for you.
- ✗ Slouch in your chair.
- ✗ Stand for a long period of time on same position.

References

1. Exercises for your back [Internet]. [cited 2021 Nov 18]. Available from: <https://www.spinesurgeoninindia.com/wp-content/uploads/2019/07/Low-back-pain-exercise-chart-English.pdf>.
2. X-plain Back Exercises. Reference Summary [Internet]. [updated 2010 Dec 16, cited 2021 Nov 18]. Available from: <https://www.ummcc.rochester.edu/medialibraries/ummccmedia/medicine/general-medicine/patientcare/documents/backexercisemedline.pdf>.
3. Low back pain. Fact sheet for patients and families [Internet]. [updated 2017 Feb, cited 2021 Nov 18]. Available from: <https://kr.ihc.com/ckr-ext/Dcmnt?ncid=51061911>.



LIVE Active

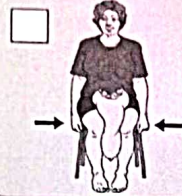
Follow these simple exercises

Knee Exercises



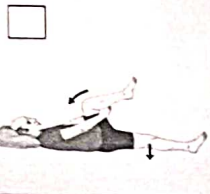
Seated Straight Leg Raise

Sit on a chair with your back straight and feet touching the floor. Lift one foot up and fully straighten the knee. Hold for a count of 5. Repeat with alternate legs.



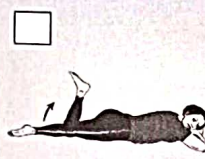
Seated Pillow Squeeze

Sit up straight on a chair with your knees bent at 90° and your feet flat on the ground. Place a pillow in between the knees and press the pillow inwards with both the knees. Hold for a few seconds and relax. Repeat this 20 times.



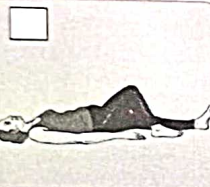
Knee to Stomach stretch

While lying on your back, pull your knee up to your stomach supporting with your hands. Push your other leg down towards the floor. Hold for a few seconds, relax and repeat with alternate legs.



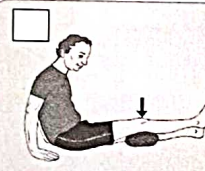
Prone Quadriceps Stretch

Lie on your stomach and bend your knee backwards towards your hip until you feel a stretch in front of the thigh. Hold for a count of 5. Repeat with both the legs 20 times each.



Supine Quadriceps stretch

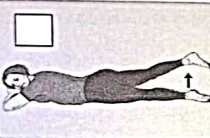
Lie on your back with legs straight. Now bring your foot closer to your hip as shown in the picture. Hold for a few seconds and relax. Repeat with alternate legs.



Quadriceps clenches

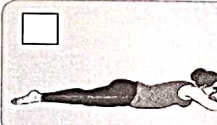
Sit with your legs straight. Keep a rolled towel under your knee. Pull the foot towards you and then press the knee against the floor while tightening the thigh muscle. Hold this for a count of five, and then fully relax. Repeat with alternate legs for 20 times.

Hip Exercises



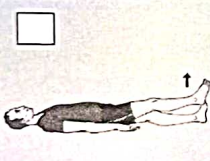
Straight-Leg Raises (Prone)

While lying on your stomach, lift your leg up by putting pressure on the thigh. Hold for a count of 5, bring the leg down and relax. Now repeat with alternate legs.



Prone Buttocks Squeeze

Lie on your stomach, squeeze or contract your buttocks. Hold for a count of 5 and relax. Repeat 10 times.



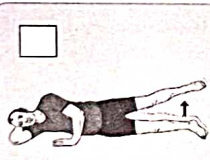
Straight-Leg Raises (Supine)

Lie on your back. Keeping the knees straight, lift alternate legs six inches off the floor, hold for a count of five and then relax. Repeat this 20 times.



Hip Bridge

Lie on your back. Bend both knees up, push on feet and lift, hold for a count of five and then relax. Repeat as comfortable.



Side Lying Leg Raise

Lie on your side with your hip on top and the bottom leg bent to provide support. Keep top leg straight and slowly lift it up as high as possible. Hold for a count of five and then relax. Repeat as comfortable on both sides.



Supine Hip Abduction

Lie on your back. Keep your toes pointed towards the ceiling. Move your leg out to the side as far as possible. Slowly return to the starting position and relax. Repeat with alternate legs.

Do's

- ✓ Exercise your arthritic joints. Regular exercise strengthens muscles and gradually reduce your pain and improve your mobility.
- ✓ Breathe normally while exercising. To avoid getting fatigued, do not hold your breath while exercising.
- ✓ Reduce your body weight through diet and exercise. Weight loss can significantly reduce stress on your joints and slow the progression of OA.
- ✓ Ask your doctor about alternate pain killer, if your pain has not reduced. Inform your physician of certain conditions that make your pain worse, such as insomnia, depression, etc.

Dont's

- ✗ Do not self medicate with pain killers, such as Diclofenac, Lornoxicam, Etoricoxib etc. Frequent and long-term use of these drugs can accelerate the progression of OA and is associated with the development of kidney and liver toxicity, causing gastric ulceration and bleeding.
- ✗ If pain increases after doing exercises, stop them and consult your doctor.
- ✗ Prolonged duration of activities that may strain or damage your joints (e.g., running, jumping, gardening, lifting heavy objects, squatting, kneeling, etc.).